Sai Manikanta Eswar Machara

+91 9392005772 | macharasaimanikantaeswar@gmail.com | Portfolio | Scholar | LinkedIn | GitHub

Research Interests

My research interests lie in *Medical Imaging* and *Computational Biology*. Beginning with Cardiac Analysis, Thyroid Nodule Segmentation, and Histopathology, my work has moved towards *Computational Biology* and *Computational Pathology*, with an emphasis on Vision-Language Segmentation, Domain Generalization across multi-centre clinical data, and multimodal risk prediction. I am drawn to problems grounded in *Clinical Reality*, where a method must hold up not only technically but where it affects patients, and I intend to pursue a PhD in Computational Biology/Pathology after my B.Tech.

Education

Rajiv Gandhi University of Knowledge Technologies (RGUKT), Nuzvid — Eluru, India
Bachelor of Technology in Computer Science and Engineering, CGPA: **9.2** — August 2023 – May 2027

Rajiv Gandhi University of Knowledge Technologies (RGUKT), Nuzvid — Eluru, India
Pre University Course, CGPA: **9.86** — December 2021 – August 2023

Experience

TANUH, Indian Institute of Science (IISc) — Bengaluru, Karnataka
Engineering Intern - Renal Health Vertical — May 2026 – Present

- Working under **Prof. Phaneendra K. Yalavarthy** (Dept. of Computational and Data Sciences) on AI-driven early detection of Chronic Kidney Disease, via a multimodal model integrating Kidney Ultrasound and Clinical parameters for automated **eGFR estimation**, CKD staging, and **KFRE-based** failure risk prediction for the Indian population.
- Developing cross-site domain generalization methods for retinal **Optical Coherence Tomography (OCT)**, so that models trained on one scanner and clinical site remain reliable on another.

Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI) — Abu Dhabi, UAE (Remote)
Research Assistant - Computational Biology — May 2026 – Present

- Working under **Prof. Yanding Zhao** on predicting **Hallmark Activity Scores** in a Pan-Cancer setting directly from H&E-stained Histopathology slides, leveraging tissue morphology to characterize underlying molecular pathway activity.
- Research spans **Prognosis prediction from Histopathology and Medical imaging** and **Single-cell Analysis**, integrating Morphological and Molecular representations to characterize Disease Progression and patient outcomes.

Khalifa University of Science and Technology — Abu Dhabi, UAE (Remote)
Research Collaborator - Medical Imaging & Computer Vision — August 2025 – Present

- Collaborating with **Dr. Iyyakutti Iyappan Ganapathi** and **Prof. Naoufel Werghi** on Vision-Language Segmentation for Histopathology and on Ultrasound-based Cardiac Phase Analysis.
- Contributed to three accepted papers from this collaboration: one at **ICCV 2025** and two at **MICCAI 2026** workshops.

Publications

Peer-Reviewed / Accepted

[1] **CIPS-Net: Text-Instructed Conditional Nucleus Instance Segmentation Network in Histopathology**
N. R. Sulake[†], **Sai Manikanta Eswar Machara**[†], S. Retta, I. I. Ganapathi, M. Owais, I. Hussain
COMPAYL Workshop, Medical Image Computing and Computer Assisted Intervention (MICCAI), 2026. [Google Scholar]

[2] **Speckle-Aware Signal Extraction as an Alternative to Complex Methods for ECG-Free Cardiac Phase Detection**
N. R. Sulake, **Sai Manikanta Eswar Machara**, S. Retta, I. I. Ganapathi, M. Owais, I. Hussain
ASMUS Workshop, Medical Image Computing and Computer Assisted Intervention (MICCAI), 2026. [Google Scholar]

[3] **DebrisVision: Bridging the Synthetic-to-Real Gap for Enhanced Underwater Debris Analysis**
S. Retta, **Sai Manikanta Eswar Machara**, I. I. Ganapathi, D. Velayudhan, N. Werghi
IEEE/CVF International Conference on Computer Vision (ICCV) Workshops, 2025. [Google Scholar]

Under Review

[4] **A Hyperbolic One-Class Fine-Tuning of Foundation Models for Robust Medical Deepfake Detection**
S. Sharma†, **Sai Manikanta Eswar Machara**†, N. R. Sulake†, S. Kethavath, P. K. Yalavarthy
IEEE Transactions on Image Processing.

Preprints & Technical Reports

[5] **GOOSE-M2F: Adapting Mask2Former for High-Fidelity, Long-Tailed Fine-Grained Semantic Segmentation in Unstructured Outdoor Terrain**
J. Lingam, N. R. Sulake, **Sai Manikanta Eswar Machara**
Technical Report, GOOSE 2D Fine-Grained Semantic Segmentation Challenge at ICRA, 2026. [Google Scholar]

[6] **MINDER: Machine Learning Framework for Depression Score Analysis in Mindfulness Interventions**
N. R. Sulake, **Sai Manikanta Eswar Machara**, D. Katam
arXiv Preprint, 2026. **(To be posted)**

† Denotes equal contribution.

Projects

RefYOLO-HuMaR: Multi-Human Referring Expression Grounding

- Constructed a Human-centric Grounding Dataset of **95K+ images** and **540K+ annotations** by integrating CrowdPose, MPII, OpenImagesV7, and COCO with automated keypoint-mask generation using ViTPose and SAM2.
- Generated instance-level referring Captions using InternVL 3.5–2B and designed a token-level Text–Visual alignment framework with MiniLM-based query encoding, bidirectional cross-attention and MIRL-based hard negative mining.
- Achieved **43.23% Margin Success Rate** on a 20K held-out subset; full-scale training and ablation studies in progress for publication.

GeoSense: Automated Rural Feature Extraction

- Developed an AI pipeline extracting rural features from drone imagery at **95.7% pixel accuracy**, with GIS-ready outputs (GeoPackage, COG).
- Designed a **two-stage model** (SegFormer-B4 + EfficientNet) for segmentation and subtype classification, achieving **IoU up to 0.92**, and handled severe class imbalance using **LDAM-DRW** and **15× class-aware sampling** (**Bridge IoU = 0.69** for rare classes).
- Processed **15K+ patches across 9 villages**, reducing manual GIS effort from **weeks to minutes** through a fully automated pipeline.

Technical Skills

Programming Languages: Python, C

Frameworks & Libraries: PyTorch, Hugging Face Transformers, Ultralytics YOLO, OpenCV, scikit-learn

Medical AI: Ultrasound & OCT Analysis, Computational Pathology (H&E, Nucleus Instance Segmentation), Clinical Risk Modelling (eGFR, KFRE), Multimodal Image–Text Fusion

Computer Vision: Text-Instructed & Vision–Language Segmentation, Long-Tailed Segmentation (LDAM-DRW, Mask2Former), Referring Expression Grounding, Synthetic-to-Real Transfer, Domain Generalization

Data & Tools: Large-Scale Dataset Construction (ViTPose + SAM2 auto-annotation, VLM captioning), Git, LaTeX, Linux

Honors & Awards

- **Runner-Up (Top 2) at AB PM-JAY Auto-Adjudication Hackathon, IISc Bengaluru**: Built **Forgensic**, a zero-model Classical CV pipeline for Document Forgery and Deepfake Detection among 3,500+ participants nationwide. **2026**
- **Top 3 at the GOOSE 2D Fine-Grained Semantic Segmentation Challenge, ICRA**: For **GOOSE-M2F**, a Mask2Former-based approach to long-tailed segmentation of unstructured outdoor terrain. **2026**
- **Finalist (Top 10, Track 1) at the IEEE EMBS BHI Data Competition**: Developed **MINDER**, an ML framework for BDI depression score prediction. **2025**
- **Winner at AI4AP Police Hackathon, Guntur**: **1st place** in **Use Case 6** (Personnel Evaluation) for an AI document analysis and visualization tool. **2025**
- **Top 140, ISRO Bharatiya Antariksha Hackathon**: Selected nationally for a hybrid deep learning model for high-resolution GIS and remote sensing imagery. **2025**
- **2nd Runner-up at IEEE INDICON Intel AI Hackathon, IIT Kharagpur**: For **CardioLens**, an echocardiogram analysis system; placed **3rd** among top research prototypes. **2024**
- **Top 22 at MAIC (Multimodal AI4TB Challenge), South Korea**: Represented **India** with a cross-modal model for TB diagnosis from X-rays and clinical notes. **2024**

Academic Service

Reviewer, MARINE Workshop 2026
European Conference on Computer Vision (ECCV)